Butterfly Effect, Chaos and Fractals: The Story Of Complex Systems

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UG, Economic Sciences @ IISER Bhopal

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So what are complex systems?

Introduction

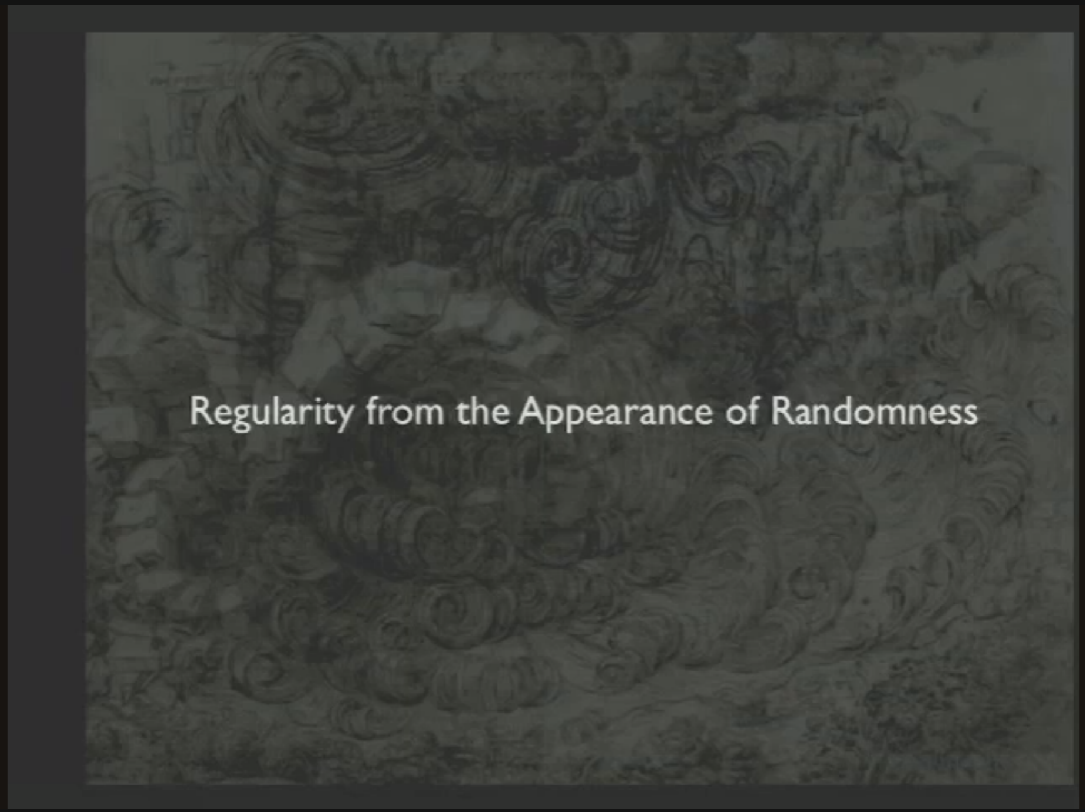
- Watery chaos? The turbulent flow of a water stream, is that complex?
- Solar system and the history of the universe, is that complex?
- the origin of life, the possible future of life, the origin of mind and the annihilation of the mind by machine, are all of these complex?
- emergent behavior and complexity are fancy words for stuff we didn't expect to happen - fundamental understanding into core dynamics of large scale phenomena like epidemics, migration, segregation, politics, cultural change, institutional change, economic collapse and financial markets

Look at this:

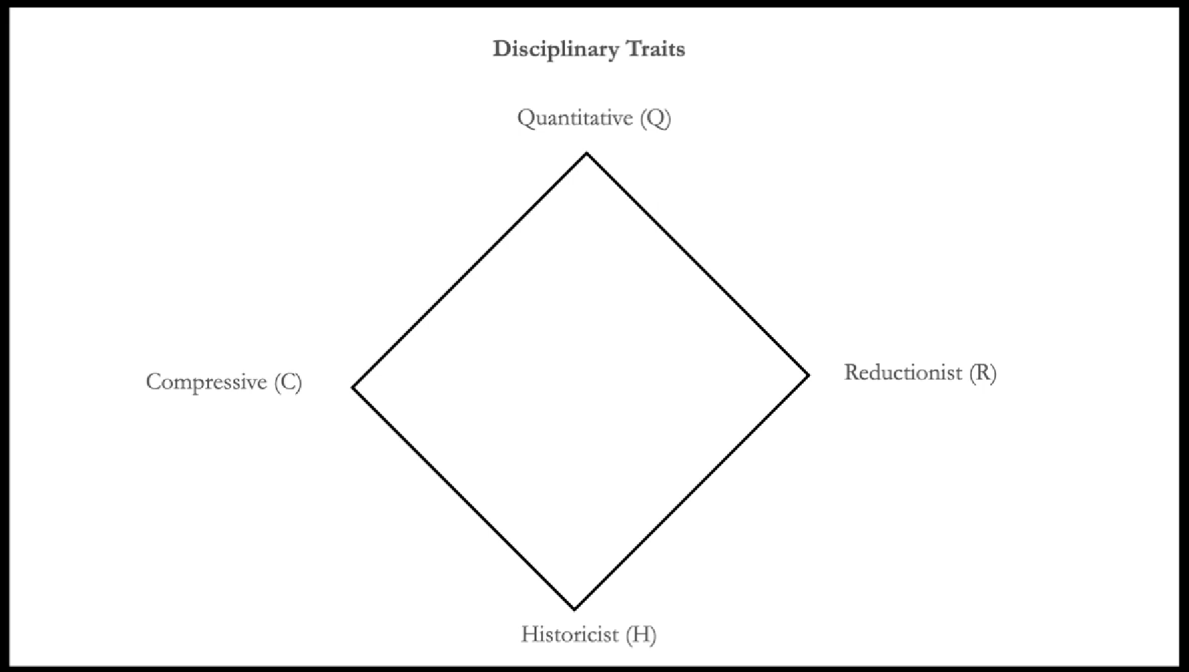
Poetry in the sky: ▸ Click Here to Watch the Video

Outbreaks: ▸ Click Here to Watch the Video

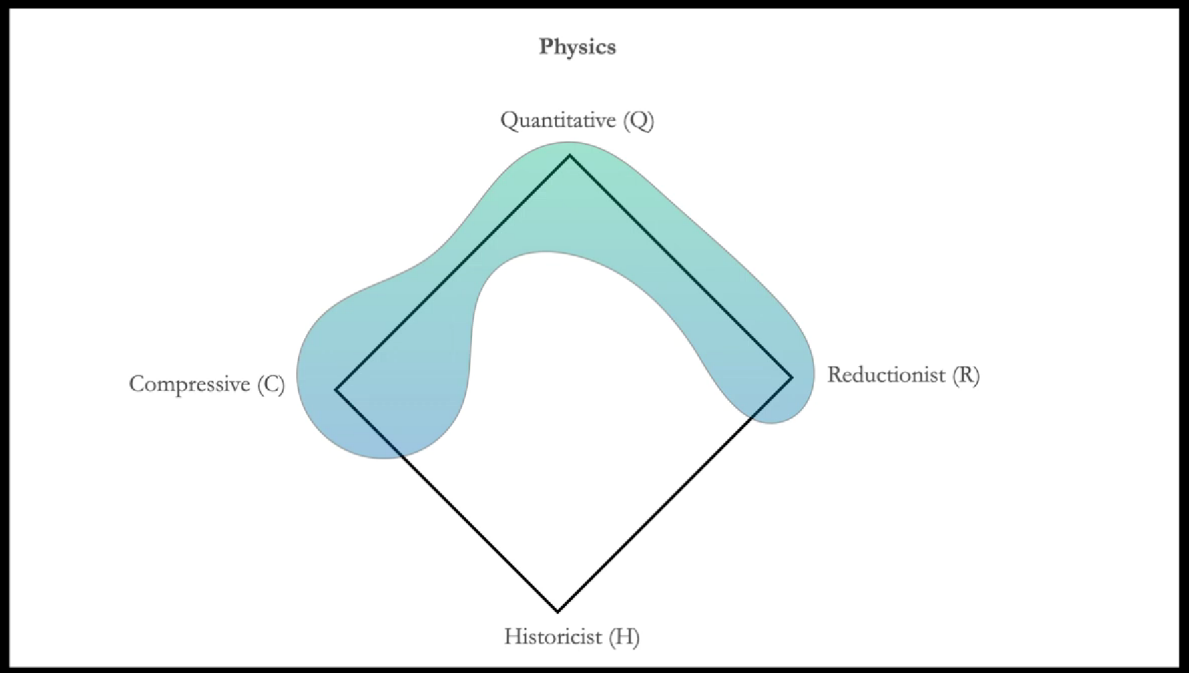
the watery chaos



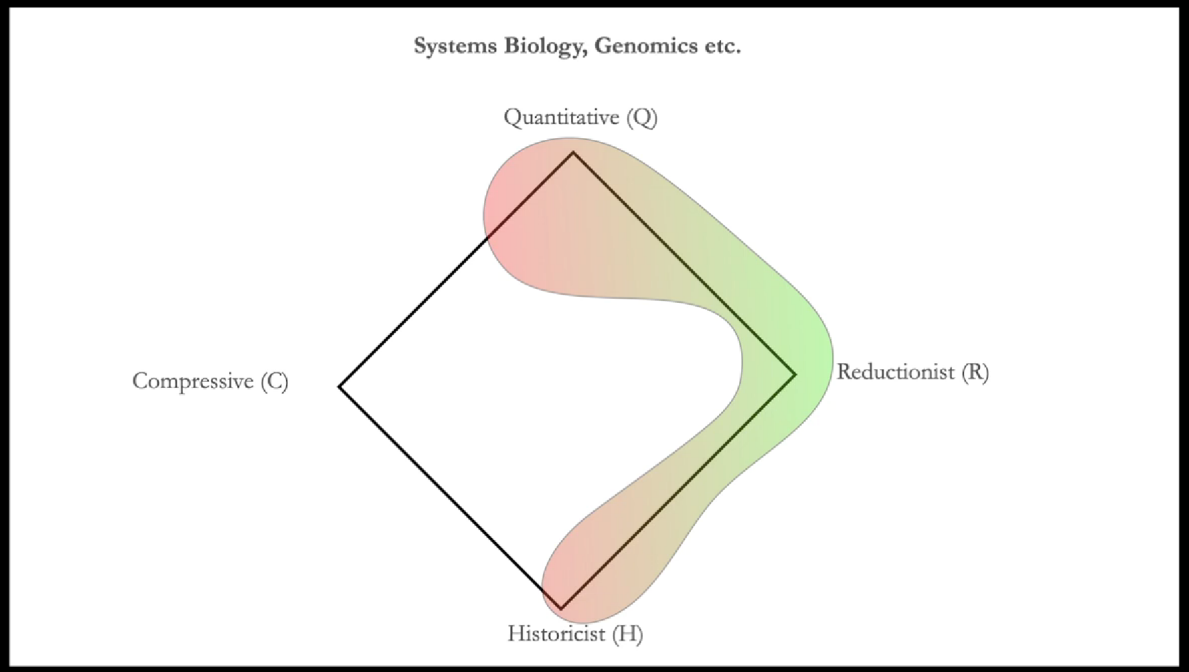
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Disciplinary Traits
Quantitative (Q)
Compressive (C)
Reductionist (R)
Historicist (H)

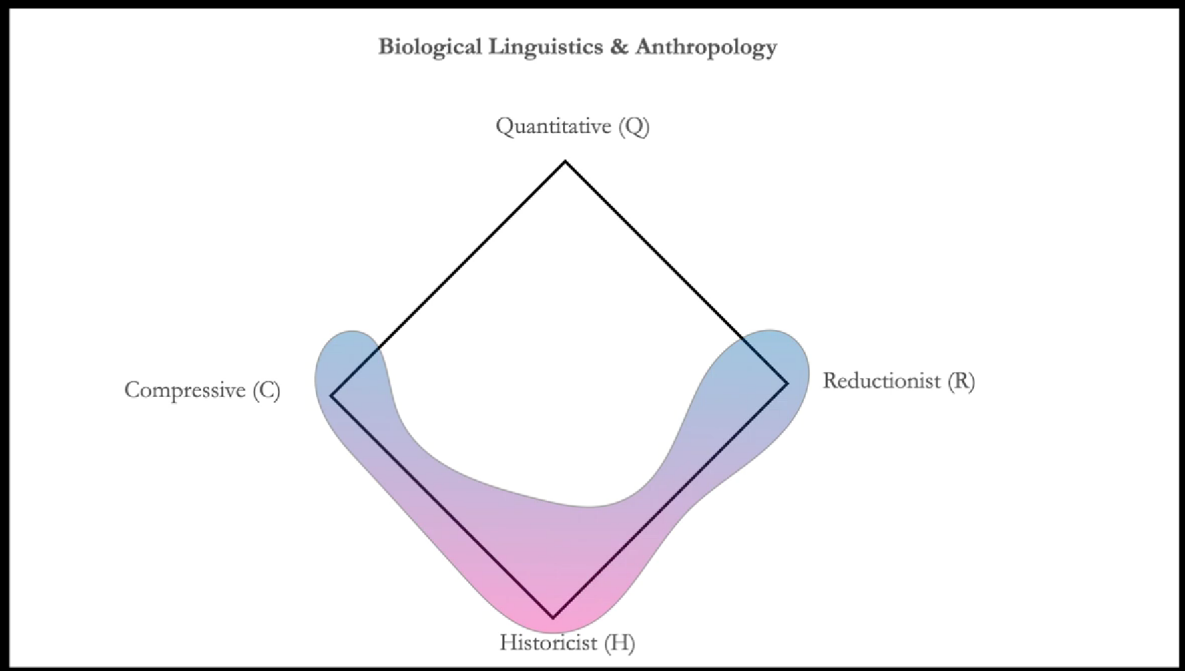
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Physics
Quantitative (Q)
Compressive (C)
Reductionist (R)
Historicist (H)

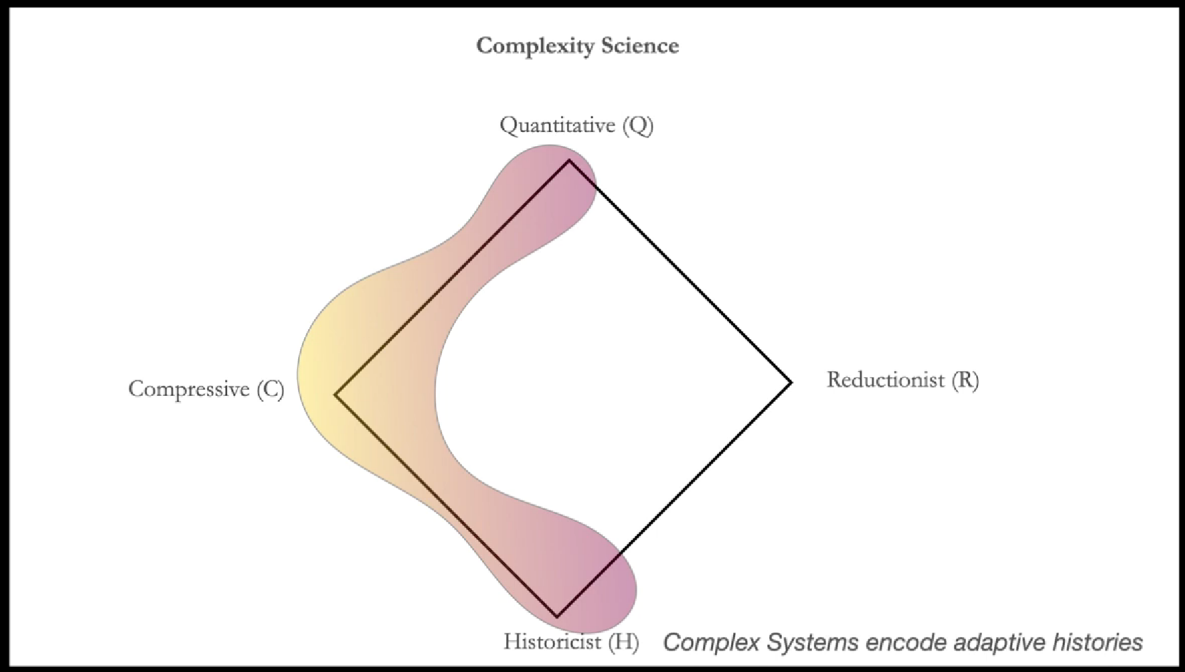
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Systems Biology, Genomics etc.
Quantitative (Q)
Compressive (C)
Reductionist (R)
Historicist (H)

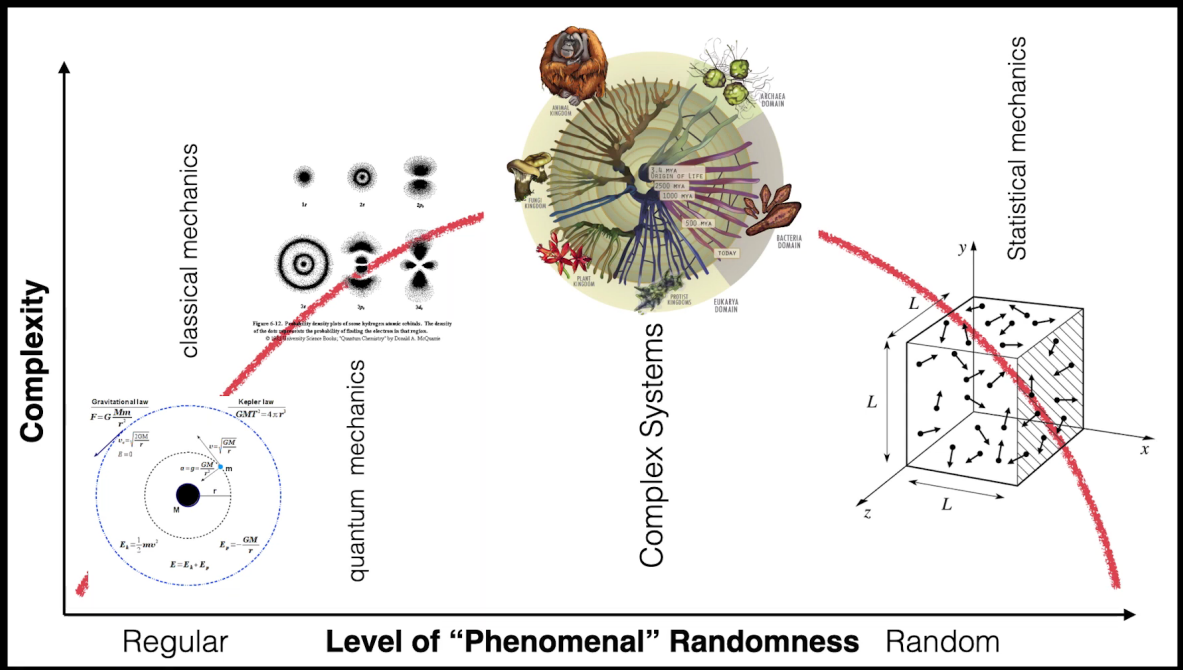
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Biological Linguistics & Anthropology
Quantitative (Q)
Compressive (C)
Reductionist (R)
Historicist (H)

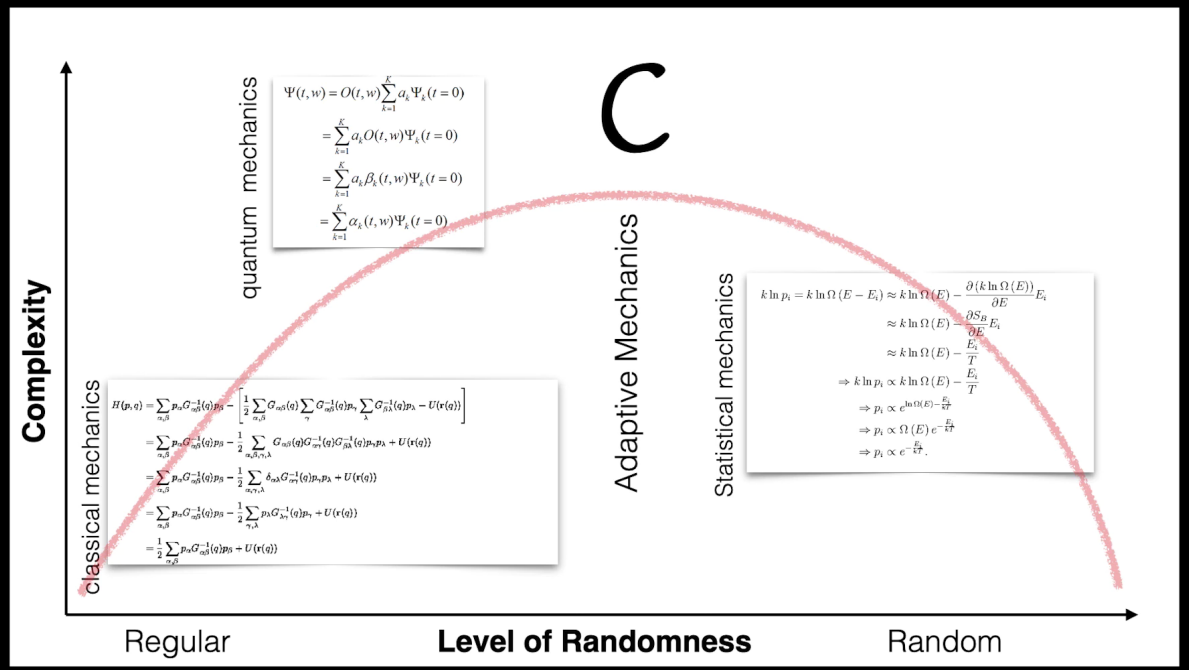
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Complexity Science
Quantitative (Q)
Compressive (C)
Reductionist (R)
Historicist (H)
Complex Systems encode adaptive histories

the complex domain



what is random and what is complex

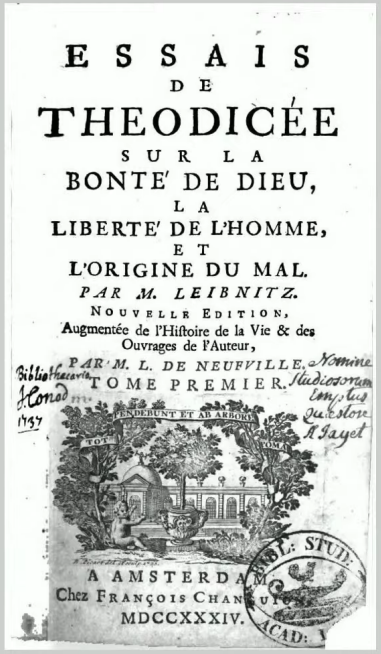


the lore

from lisbon to leibniz

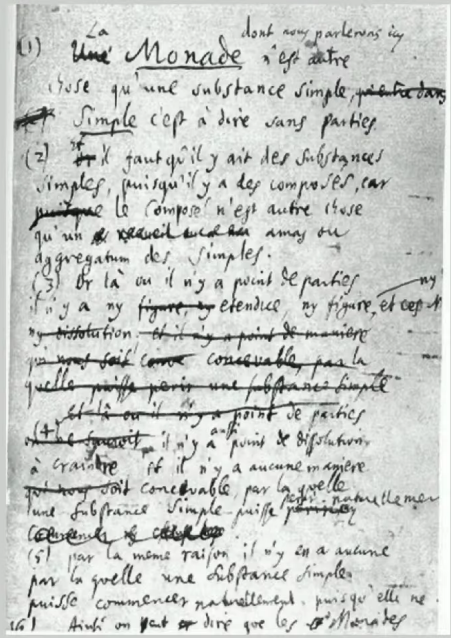


from lisbon to leibniz



ESSAIS
DE
THEODICÉE
SUR LA
BONTE' DE DIEU,
LA
LIBERTE' DE L'HOMME,
ET
L'ORIGINE DU MAL.
PAR M. LEIBNITZ.
NOUVELLE EDITION,
Augmentée de l'Hiſtoire de la Vie & des Ouvrages de l'Auteur,
PAR M. L. DE NEUFVILLE.
TOME PREMIER.
A AMSTERDAM,
Chez FRANÇOIS CHANGUION.
MDCCXXXIV.

Gottfried Leibniz, 1710 (1735)



(1) La Monade dont nous parlerons ici
n'est autre
chose qu'une substance simple,
Simple c'est à dire sans parties
(2) Il faut qu'il y ait des substances
simples, puisqu'il y a des composés, car
le composé n'est autre chose
qu'un amas ou
aggregatum des simples.
(3) Or là où il n'y a point de parties
ny
il n'y a ny figure, ny etendue, ny
de maniere
conceuable, par la
quelle puisse perir une substance simple
et là où il n'y a point de parties
il n'y a aussi point de dissolution.
à craindre et il n'y a aucune maniere
conceuable par la quelle
une substance simple puisse
naturellement
(5) par la même raison il n'y en a aucune
par la quelle une substance simple
puisse commencer naturellement, puisqu'elle ne

Gottfried Leibniz, 1714

the city

Scaling

- Biological and urban systems do not scale in a purely multiplicative way, meaning that a simple multiplication of the results obtained by the prototype (a mouse or a small city) will not allow us to obtain the desired effect in another scale of interest (a human body or a large city).
- In the case of a city, we have a similar situation when we analyze many urban metrics as a function of the city population size.
- These empirical facts suggest that, despite the remarkable complexity underlying biological and urban phenomena, the relationships between metabolic rate and body mass, and between GDP and city population size seem to obey a relatively simple relationship, given by the power-law equation

what??

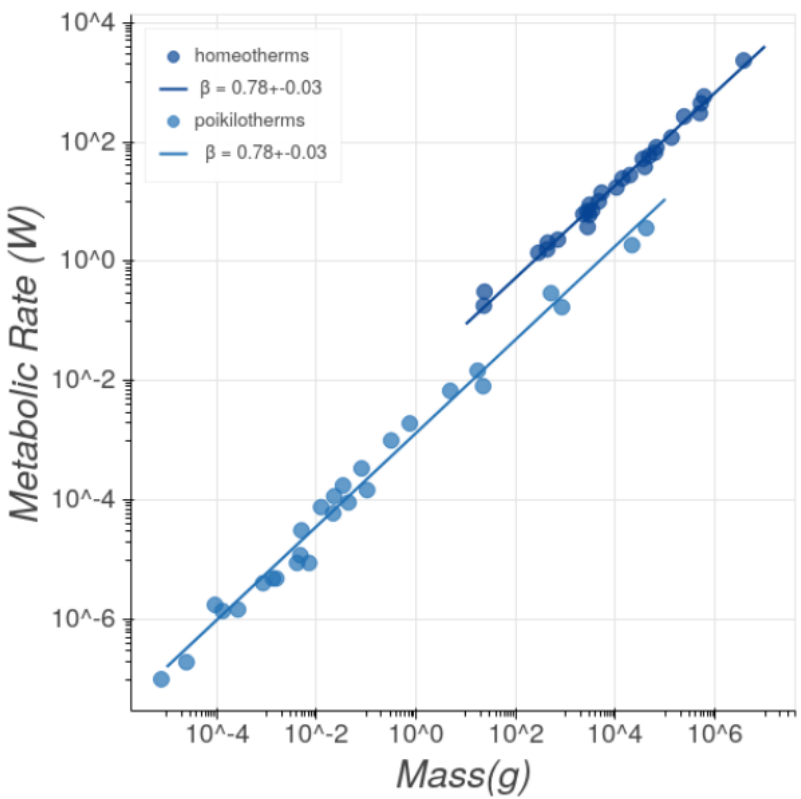


Fig. 1 Metabolic rate as a function of body mass for two vascular taxonomic groups: homeotherms (organisms that maintain a constant internal body temperature) and poikilotherms (organisms that vary their internal body temperature). A power-law of the type given by Eq. (1) is observed for all taxonomic groups (the straight lines capture the trend of points). The parameter Y_0 (the intercept) varies from group to group, but β is approximately constant and sub-linear ($\beta < 1$). The data were extracted directly from the references [4] (homeotherms) and [5] (homeotherms and poikilotherms). Source: [6, Fig.1].

what??

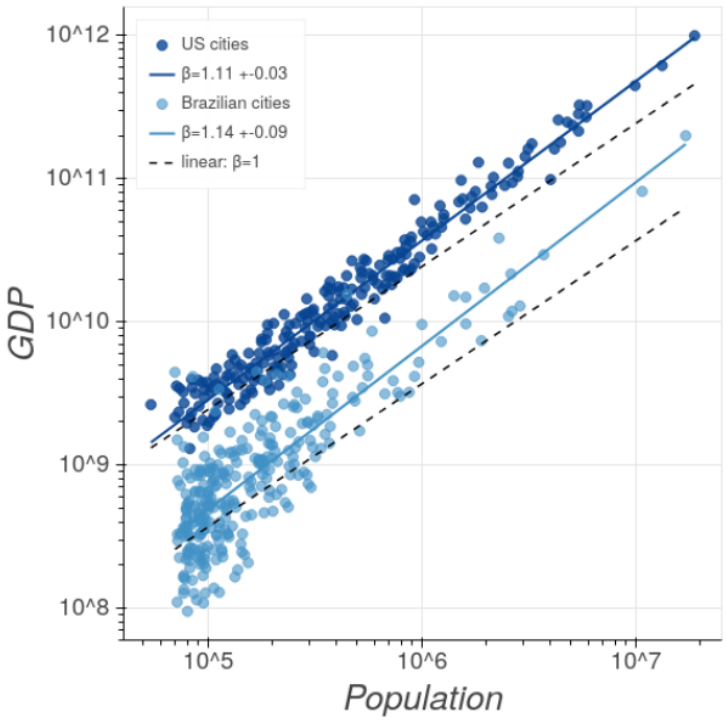


Fig. 2 Example of urban scaling. The urban gross domestic product (GDP) is plotted as a function of population N for cities in Brazil and the United States (on a double logarithmic scale). The solid lines (blue and green) are the best fits that capture the trend of the points and reveal the power-law property. Both countries exhibit scaling exponents $\beta > 1$ (super-linear), with very similar values despite the socio-economic differences between the two countries, which justifies the difference in the vertical position of the straight lines (different intercepts Y_0). The dashed lines indicate linearity ($\beta = 1$) and are included only to highlight the super-linear behavior of the data trend. Source: [12, Fig.1].

what??

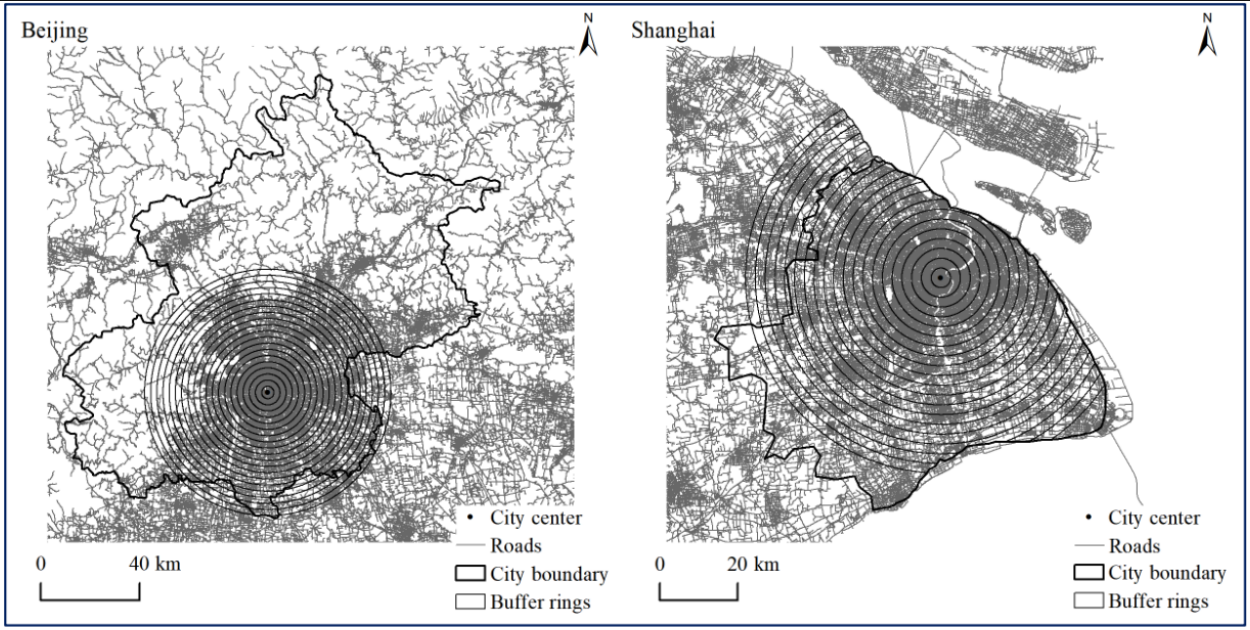


Figure 2 Sketch maps of cluster growing method for radial dimension measurement of urban traffic networks

Thank You

Questions?